

A Bilevel Hybrid Genetic Algorithm for Capacitated Electric Vehicle Routing Problem

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Abstract—As electric vehicles become more prevalent, a novel vehicle routing problem (VRP) has emerged, known as the capacitated electric VRP (CEVRP). CEVRP requires determining not only the service order of customers but also the charging plans for vehicles, thereby increasing the complexity of solution construction. In response to this challenge, we propose a bilevel hybrid genetic algorithm (BHGA). BHGA models CEVRP as two levels of subproblem: 1) the upper level capacitated VRP, focusing on the service order and 2) the lower level fixed route vehicle charging problem, focusing on the charging plans. In dealing with the upper level subproblem, the hybrid genetic search algorithm is adopted to construct the routes to visit customers and an advanced screening strategy is proposed to optimize the local search process and effectively guide the evolution of population. For the lower level subproblem, an efficient heuristic method called focus enumeration is designed, which is specifically used to insert charging stations into routes to ensure battery constraint. The collaboration of the advanced screening strategy and the focus enumeration assists in more unified solving of the two subproblems. The experiments show that BHGA significantly surpasses state-of-the-art algorithms on benchmark instances and has successfully updated eleven best known solutions, demonstrating its outstanding performance.

Index Terms—Hybrid genetic algorithm, local search, capacitated electric vehicle routing problem, combinatorial optimization.

I. INTRODUCTION

The rapid development of the transportation industry has led to significant global consumption of fossil fuels, exacerbating environmental pollution and intensifying the strain on fuel resources [1]. Electric Vehicles (EVs), due to their enormous potential in energy conservation, environmental protection, and carbon emission reduction, are now seen as a key direction for the future development of transportation [2]. With the continuous progress in electric vehicle technology, governments and

businesses worldwide have introduced numerous supportive policies, promoting the rapid growth of the electric vehicle industry [3]. This has opened up vast market opportunities for EVs in the logistics field, but it also brings new risks and challenges.

Since EVs are widely used in transportation, a new vehicle routing problem (VRP) called electric VRP (EVRP) emerges [4]. Different from the traditional VRPs, since charging is not as convenient as refueling, we need to consider the cruising range of EVs and make charging plans so that EVs will not run out of battery. Several EVRP variants have been proposed with different settings such as non-linear charging and heterogeneous vehicles [5]. Among these EVRPs, this paper focuses on the capacitated EVRP (CEVRP), which is the fundamental problem of other variants [6]. The goal of CEVRP is to minimize the total travel distance of EVs. EVs need to plan routes rationally to determine the sequence of visiting customers and charging stations. EVs leave the depot fully charged. During the journey, the battery consumption rate of EVs is constant, and they must meet capacity and battery limitations. To ensure sufficient battery, EVs may need to visit charging stations, assumed to be fully charged each time. Each customer can only be serviced by one EV. After completing the service, all EVs must return to the depot.

Currently, the methods proposed to solve routing problems can be classified into three categories: exact methods, meta-heuristic methods, and hyperheuristic methods. The application of exact methods is relatively limited in this problem, as they may encounter problems of high computational complexity and slow convergence [7]. Although attempts have been made to use the branch and bound method, its high computational complexity makes it difficult to effectively handle large-scale problems [8]. Metaheuristic algorithms show great advantages in solving such combinatorial optimization problems, providing approximate optimal solutions in a relatively short time. Therefore, they are widely used in solving CEVRP, such as ant colony optimization (ACO) [9], [10], iterative local

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search (ILS) [11], genetic algorithms (GA) [12], [13], simulated annealing (SA) [14], and variable neighborhood search (VNS) [15]. Despite the good global search performance of metaheuristic algorithms, their convergence speed can also be unsatisfactory in large-scale problem instances, requiring longer computation times to find a good solution [10], [16]. Hyperheuristic methods operate by selecting, combining, and adapting from a range of low-level heuristics, but the process of picking the appropriate heuristics can be quite challenging [17].

CEVRP can be seen as an extension of CVRP by adding the charging constraint. So far, the hybrid genetic search (HGS) algorithm has emerged as a standout for solving CVRP, distinguished by its superior solution quality and faster convergence [18]. Considering the drawbacks of the existing algorithms and the advantages of HGS, we propose a bilevel hybrid genetic algorithm (BHGA) based on HGS in this paper to solve CEVRP efficiently and effectively. BHGA divides CEVRP into two levels. The subproblem at the upper level is CVRP, temporarily ignoring the battery constraint. Then, for the subproblem at the lower level, i.e., the fixed route vehicle charging problem (FRVCP), a charging function is used to transform the solution from the upper level into a solution for CEVRP, ensuring compliance with the battery constraint. The contributions of BHGA are as follows:

- For the upper level subproblem, we propose an advanced screening (AS) strategy. For small-scale instances, a complete AS strategy is used during the local search process to ensure convergence to the best solution. For larger-scale instances, a partial AS strategy is employed, helping to explore more of the CVRP suboptimal solution space and improving the solution quality and robustness of the algorithm.
- For the lower level subproblem, we propose a new algorithm to generate the charging plans called focus enumeration (FE). FE concentrates on the focal area between the list of charging stations and route intervals. Through a pre-processing step and targeted regional enumeration, this algorithm can provide high-quality solutions efficiently for FRVCP.

The rest of this article is arranged as follows. Section II reviews the related works. Section III describes CEVRP. Section IV presents the details of BHGA. Section V describes the experimental setup. Experiments are conducted and analyzed in section VI. Finally, section VII draws the conclusion.

II. RELATED WORKS

In this section, we delve into the methods for solving EVRPs, with a particular focus on the CEVRP. Currently, the primary approaches to solving EVRPs include exact methods, metaheuristic methods, and hyperheuristic methods.

Given the scale and complexity of EVRPs, the application of exact methods in EVRPs research is relatively limited. Commercial solvers such as CPLEX and Gurobi have been used to solve EVRPs. CPLEX was employed to solve EVRP with time windows (EVRPTW) and stochastic waiting times at

charging stations [19]. Gurobi was used in solving EVRPTW with battery swapping stations [20]. These experimental results indicate that while exact methods can provide high-quality results for small-scale instances, they face challenges of computational complexity in large-scale problems. Moreover, variants of EVRP include more complex constraints, such as time windows and multiple vehicle types, which may further impede the application of exact methods.

In CEVRP, metaheuristic methods are widely applied due to their superior efficiency. The IEEE WCCI2020 competition of EVRP introduced a benchmark data set [6], where instances ranged from 21 to 1000 customers, facilitating comparisons of algorithms. At that time, Woller's VNS algorithm and Mak-Hau's SA algorithm demonstrated exceptional performance across different scale instances, securing the first and second places in the competition respectively [21]. Moreover, Hien *et al.* [22] introduced a clustering-inspired greedy search algorithm (GS) integrated into genetic algorithm, creating GSGA, which won third place in the competition. Woller also proposed an algorithm called GRASP [23], which proved to be quite effective. Jia *et al.* [10] modeled CEVRP as a bilevel optimization problem, with the upper level being CVRP and the lower level being FRVCP, and introduced a bilevel ACO (BACO) algorithm. Specifically, the upper level subproblem was solved using the order-first, split-second max-min ant system (OS-MMAS) algorithm, and the lower level subproblem utilized a removal heuristic (RH) and a restricted enumeration method (RE) for determining charging sequences. The BACO algorithm updated multiple best known solutions, exhibiting superior performance over VNS and SA in a wide range of instances. Moreover, they proposed a confidence-based ACO (CBACO) [16], adding a confidence-based selection method between the two levels of subproblems, enhancing the solution quality and efficiency of the algorithm. Lin *et al.* [24] introduced an extended ACO (EACO) algorithm, employing a fine route construction method to generate sequences of customers with charging stations. Wang *et al.* [25] proposed the dual-population-based co-evolutionary algorithm (DPCA), employing two populations for collaborative optimization of route planning and charging decision problems.

Furthermore, hyperheuristic methods also emerge as a feasible way to solve CEVRP. Rodriguez *et al.* [17] introduced a hyper-heuristic adaptive simulated annealing and reinforcement learning (HHASA-RL) algorithm, providing some competitive results for the WCCI2020 competition of EVRP.

Relying on appropriately configured search mechanisms, although metaheuristic methods represented by CBACO and hyperheuristic methods represented by HHASA-RL have achieved commendable results in CEVRP, there is still room for improvement in terms of convergence speed and solution quality in large-scale instances. Therefore, this paper proposes BHGA for further optimization.

III. PROBLEM DEFINITION

CEVRP is an extension of CVRP and can be represented by a fully connected weighted graph $G = (V, E)$. $V = \{0\} \cup V_c \cup$

V_f is the set of nodes, and E is the set of edges connecting these nodes. $\{0\}$ represents the depot, from which EVs depart, provide services to customers in sequence, complete charging as needed, and then return to the depot. V_c represents the set of customers. The positive freight demand of each customer $i \in V_c$ is denoted as c_i . V_f represents charging stations, with EVs being fully charged when leaving the charging station and depot. For formulation, $\hat{V}_f$ is defined as an expansion of V_f , containing $2|V_c|$ copies of each charging station. For each EV, the load limit is denoted as C , and the battery limit is denoted as Q . During the journey of each EV, the total demand of customers should not exceed the load limit C , and the current consumed power should not exceed the battery limit Q . Each edge $(i, j) \in E$ is associated with a non-negative distance d_{ij} , representing the distance between node i and j . The battery consumption rate is represented by a constant h , meaning that after passing each edge (i, j) , the EV will consume $h \cdot d_{ij}$ amount of electricity. The mathematical definition of CEVRP is as follows [6]:

$$\min f(\mathbf{x}) = \sum_{i \in V, j \in V, i \neq j} d_{ij} x_{ij}, \quad (1)$$

$$\text{s.t.} \quad \sum_{j \in V, i \neq j} x_{ij} = 1, \forall i \in V_c, \quad (2)$$

$$\sum_{j \in V, i \neq j} x_{ij} \leq 1, \forall i \in \hat{V}_f, \quad (3)$$

$$\sum_{j \in V, i \neq j} x_{ij} - \sum_{j \in V, i \neq j} x_{ji} = 0, \forall i \in V, \quad (4)$$

$$u_j \leq u_i - c_i x_{ij} + C(1 - x_{ij}), \forall i \in V, \forall j \in V, i \neq j, \quad (5)$$

$$0 \leq u_i \leq C, \forall i \in V, \quad (6)$$

$$y_j \leq y_i - h d_{ij} x_{ij} + Q(1 - x_{ij}), \forall i \in V_c, \forall j \in V, i \neq j, \quad (7)$$

$$y_j \leq Q - h d_{ij} x_{ij}, \forall i \in \hat{V}_f \cup \{0\}, \forall j \in V, i \neq j, \quad (8)$$

$$0 \leq y_i \leq Q, \forall i \in V, \quad (9)$$

$$x_{ij} \in \{0, 1\}, \forall i \in V, \forall j \in V, i \neq j. \quad (10)$$

Equation (1) indicates that the goal of CEVRP is to minimize the total travel distance of all EVs. Equation (2) stipulates that each customer receives service only once. Equation (3) allows a charging station to be visited multiple times. Considering the worst case that one EV needs to visit a charging station before and after a service, it is completely sufficient to have $2|V_c|$ copies of each charging station. Equation (4) ensures that each customer receives the corresponding service. Equations (5) and (6) are capacity constraint, requiring that EVs should not exceed their load limit when reaching any node. Equations (7), (8), and (9) are battery constraint, stipulating that EVs should not run out of power during the journey. Equation (10) defines a set of boolean variables, indicating whether EVs travel along edge (i, j) .

IV. PROPOSED METHOD

A. Overall Process of BHGA

BHGA divides CEVRP into two layers, upper and lower, and then obtains solutions that meet the constraint conditions.

Algorithm 1: BHGA

Input: CEVRP instance,
 $It_{\max}$: iterative stability coefficient,
 $T_{\max}$: maximum execution time,
 $step$: step that constrain AS in local search,
 λ : penalty coefficient

Output: $\mathbf{x}_{gb}$: global best solution

- 1 initialize population Π with random solutions improved by *localSearchWithAS*;
- 2 **while** number of unimproved iterations $< It_{\max}$ and CPU time $< T_{\max}$ **do**
- 3 $P_1, P_2 = \text{tournamentSelection}(\Pi)$;
- 4 $C_1 = \text{orderCrossover}(P_1, P_2)$;
- 5 $C_2 = \text{localSearchWithAS}(C_1, step, \lambda)$;
- 6 $C_e = C_2$;
- 7 **if** $\text{cost}(C_1) < \text{cost}(C_e)$ **then**
- 8 $C_e = C_1$;
- 9 **end**
- 10 insert C_e into respective subpopulation;
- 11 **if** C_e is infeasible **then**
- 12 C_e has 50% probability of being improved by *localSearchWithAS* and inserted into respective subpopulation;
- 13 **end**
- 14 **if** $\text{size}(\text{subpopulation}) > \text{limit}$ **then**
- 15 select survivors in subpopulation;
- 16 **end**
- 17 adjust λ for infeasibility;
- 18 **end**
- 19 select the best feasible solution from Π as $\mathbf{x}_{gb}$;
- 20 **return** $\mathbf{x}_{gb}$;

The subproblem of the upper layer is CVRP, temporarily ignoring the battery constraint. For the sub-problem of the lower layer, i.e. FRVCP, we propose a heuristic called FE as the charging function to insert charging stations into the solutions of the upper layer. After obtaining the whole solution, we use the evaluation function to assess the individuals. Finally, the best solution is updated if the new solution is better.

The overall process of BHGA is shown in Algorithm 1. First, we randomly initialize the population Π and improve it using local search with AS (line 1). Based on whether the capacity constraint is satisfied, we divide Π into two subpopulations: feasible and infeasible solutions. Then, BHGA begins the iterative process (line 2). Binary tournament selection is applied to the union of the two subpopulations (line 3), randomly selecting two individuals and retaining the one with better fitness as the parent. We use P_1 and P_2 to represent the parents obtained through binary tournament selection. P_1 and P_2 produce a new individual C_1 through ordered crossover (line 4). We apply local search with AS to C_1 , and the individual obtained is denoted as C_2 (line 5). The algorithm retains one of C_1 and C_2 with better total CEVRP travel distance performance as C_e (lines 6-9). According to its feasibility, we add C_e to the corresponding subpopulation (line 10). If C_e is infeasible, the algorithm will also improve it with a 50% probability using local search with AS at a 10 times penalty score, and the improved result is added to the corresponding subpopulation (lines 11-13). When adding

a new individual to Π , if the size of subpopulation exceeds the limit, survivor selection will be triggered, prioritizing the elimination of clone individuals, followed by the individuals with the worst fitness (lines 14-16). Then, we adjust the penalty coefficient λ (line 17) as mentioned in [18]. Finally, when the number of unimproved iterations or CPU time reaches the iteration condition, BHGA will return the best solution it ever found.

B. Solution Representation

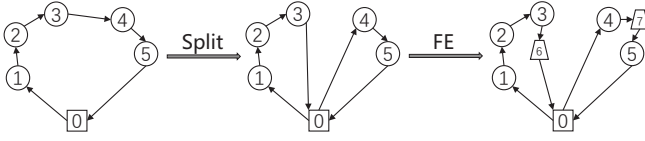


Fig. 1. Solution representation in BHGA.

As illustrated in Fig. 1, in BHGA, chromosome of individual is represented as a giant TSP sequence. This design is more effective in protecting genetic information during crossover. Prins [26] proposes a Split algorithm to split a TSP sequence into several routes that meet the capacity constraint, replacing trip delimiters. For computational efficiency, BHGA uses a faster split algorithm called linear-time Split proposed by [27]. For each route, we use FE to handle the battery constraint by making a reasonable charging plan. Finally, a CEVRP solution is generated.

Following HGS [18], the fitness of an individual in BHGA is defined as:

$$f(I) = rank_{\phi}(I) + (1 - \frac{n_{elite}}{|\Pi|})rank_{div}(I). \quad (11)$$

It is designed based on two main factors: the ranking of solution quality $rank_{\phi}(I)$ and the ranking of diversity contribution $rank_{div}(I)$. Here, the diversity contribution is determined by calculating the average broken-pairs distance between the individual and its several most similar individuals. Furthermore, n_{elite} represents the number of individuals with the best performance in Π .

C. Local Search with Advanced Screening

In BHGA, we propose an AS strategy to enhance the local search moves, as shown in Algorithm 2. For the CVRP routes of an individual, u is randomly selected from all nodes, and v is selected from a specific neighborhood of u (line 1). Each local search iteration traverses all possible node pairs and tries the following ten local search moves in turn (line 3): 1) six basic moves, 2) three 2-opt moves, and 3) a swap* move [18]. The swap* move seeks the optimal swap of two nodes between $r(u)$ and $r(v)$, allowing the nodes after the exchange to be placed at any appropriate position in the routes.

For the local search moves, pre-pruning is necessary. To help the population evolve better towards CEVRP, BHGA introduces a strategy. Considering after CVRP pre-pruning, there are two scenarios: CVRP solutions become better or

Algorithm 2: Local Search with AS

Input: I : individual,
 $step$: step of AS,
 λ : penalty coefficient,
 FE : function to insert charging stations,
 $overLoadPenalty$: function to calculate penalty when the route is overloaded

Output: I^* : improved individual

```

1 initialize the nodePairs according to  $I$ , then shuffle the order;
2 for  $(u, v)$  in nodePairs do
3   for move in moves do
4     call the corresponding move function;
5     Function move ( $u, v, step, \lambda$ ):
6        $r(u)$  denotes the route where node  $u$  is located;
7        $r(v)$  denotes the route where node  $v$  is located;
8        $\Delta r(u)$  and  $\Delta r(v)$  denote the CVRP increment in route resulting from the move;
9        $\Delta r(u) += overLoadPenalty(r(u), \lambda)$ ;
10       $\Delta r(v) += overLoadPenalty(r(v), \lambda)$ ;
11      if  $\Delta r(u) + \Delta r(v) > 0$  then
12        return;
13      end
14      if the current move is within the range of step then
15         $\Delta' r(u)$  and  $\Delta' r(v)$  denote the CEVRP increment in route resulting from the move, calculated by  $FE$ ;
16        if  $\Delta' r(u) + \Delta' r(v) > 0$  then
17          return;
18        end
19      end
20      modify  $r(u)$  and  $r(v)$  accordance with the current move and update the relevant information;
21    End Function;
22    if  $r(u)$  or  $r(v)$  has been modified then
23      continue;
24    end
25  end
26 end
27 return  $I^*$ ;

```

remain unchanged. Since CVRP solutions are the basis of CEVRP solutions, and high-quality CVRP solutions help solve CEVRP, the search space after CVRP solutions improve is obviously more exploratory. When observing VRP solution improvements (lines 11-13), we apply the charging function to the routes (line 15). If the power is feasible, assess whether the CEVRP solution has improved (lines 16-18). If there is an improvement, retain this modification (line 20). Moreover, the parameter $step$ indicates that AS will be applied to the first $step$ moves of local search (line 14).

D. Focus Enumeration

The FE algorithm is designed for FRVCP, aiming to reasonably insert charging stations into CVRP routes to meet the battery constraint. Two main features distinguish FE from other heuristic algorithms for charging: 1) vital charging stations pre-processing and 2) accurate regional enumeration.

Algorithm 3: FE

Input: r : an electricity-infeasible route,
 $G = (V, E)$: $V = \{0\} \cup V_c \cup V_f$, E is edge set,
 X : A matrix designed to store the advantageous
charging stations between any two nodes in $\{0\} \cup V_c$,
 Q : maximum battery capacity,
 h : battery consumption rate
Output: r^* : an electricity-feasible route

```

1 initialization is required only once in each instance;
2 foreach  $(0, p), p \in V_c$  do
3   | add charging stations with non-surpass relationship to
   |  $X_{0p}$  and  $X_{p0}$ ;
4 end
5 foreach  $(i, j), (i, j \in V_c) \wedge (i < j)$  do
6   | add the nearest charging station to  $X_{ij}$  and  $X_{ji}$ ;
7 end
8  $d_r$  denote the CVRP travel distance of  $r$ ;
9  $maxdis = Q/h$ ;
10  $lb = \lfloor d_r / maxdis \rfloor$ ;
11  $ub = lb + 1$ ;
12 for  $sn = lb \rightarrow ub$  do
13   | call the following recursion function;
14   | Function routeCharge( $r, r_{sn}, tmp, sn, n = sn$ ):
15   |   | if  $n == 0$  then
16   |   |   | based on  $tmp$ , attempt to update  $r_{sn}$ ;
17   |   |   | return;
18   |   | end
19   |   | for interval( $i, j$ ) in  $r$  do
20   |   |   | check whether the route before the current
   |   |   | interval satisfies the battery constraint;
21   |   |   | evaluate whether the EV can successfully
   |   |   | complete its journey back to depot based on
   |   |   | the remaining number of charging times;
22   |   |   | for  $cs$  in  $X_{ij}$  do
23   |   |   |   | check whether the previous journey satisfies
   |   |   |   | the battery constraint after choosing  $cs$ ;
24   |   |   |   | evaluate whether the EV can successfully
   |   |   |   | complete its journey back to depot based
   |   |   |   | on the remaining number of charging times
   |   |   |   | after choosing  $cs$ ;
25   |   |   |   | if all the previous conditions are met then
26   |   |   |   |   | choose  $cs$  as the charging station, and
   |   |   |   |   | save relevant information to  $tmp$ ;
27   |   |   |   | end
28   |   |   |   | routeCharge( $r, r_{sn}, tmp, sn, n - 1$ );
29   |   |   | end
30   |   | end
31   | End Function;
32   | based on  $r_{sn}$ , attempt to update  $r^*$ ;
33 end
34 return  $r^*$ ;

```

The pre-processing stage filters charging stations with advantages. When dealing with charging stations between customers, we prioritize the single charging station that brings the least additional driving distance (lines 5-7). In fact, during the later stages of population optimization, the distances between customers are usually shorter, hence choosing a single charging station is more cost-effective. In contrast, when addressing the charging stations between depot and customer, we focus on multiple charging stations that possess a “non-surpass” relationship (lines 2-4). The logic is that the EV is fully

TABLE I
CEVRP BENCHMARK INSTANCES

Name	Customer	Station	Route
E22	21	8	4
E23	22	9	3
E30	29	6	4
E33	32	6	4
E51	50	5	5
E76	75	7	7
E101	100	9	8
X143	142	4	7
X214	213	9	11
X351	350	35	40
X459	458	20	26
X573	572	6	30
X685	684	25	75
X749	748	30	98
X819	818	25	171
X916	915	9	207
X1001	1000	9	43

charged when leaving depot, so under the premise of the same additional driving distance, the charging station farther from depot should be prioritized. We define the “surpass” relationship as follows:

Definition 1. Given the depot 0 and $p \in V_c$, a station s_1 is said to surpass another station s_2 , if and only if

$$[(d_{0,s_1} + d_{s_1,p}) < (d_{0,s_2} + d_{s_2,p})] \wedge (d_{s_1,p} < d_{s_2,p}) \quad (12)$$

The second focus of the algorithm is on enumerating promising regions, mainly achieved through retrospective check and prospective pruning. The algorithm traverses the intervals of the route and attempts to insert charging stations, a process similar to depth-first search. Retrospective check ensures that the previous part of the route meets the battery constraint (line 20, 23); If not, it can be judged that the current charging scheme is infeasible, and the algorithm will stop the current branch search and turn to other branches. On the other hand, prospective pruning then focuses on the electrical feasibility of the remaining part of the route (line 21, 24). This dual check mechanism ensures that the algorithm only enumerates the charging station combination schemes of important intervals, avoiding ineffective searches.

Through the above two focal strategies, the FE algorithm recursively attempts to insert charging stations into the route and evaluates the feasibility of the route after each attempt (lines 19-30). If a feasible route is obtained, its cost is compared with the current best solution. If the new route is better, it is updated as the current best solution (line 32).

V. EXPERIMENTAL SETUP

A. Benchmark Instances

The WCCI2020 competition of EVRP proposed a benchmark that includes 7 small-scale instances and 10 large-scale instances. This benchmark is a classic test dataset for the CEVRP. The details of these instances are shown in Table I. Several advanced algorithms have been tested on this benchmark, allowing for a relatively fair comparison.

TABLE II
COMPARISON BETWEEN BHGA AND THE STATE-OF-THE-ART ALGORITHMS ON 17 BENCHMARK INSTANCES

case	index	BKS	GSGA	VNS	BACO	CBACO-I	HHASA-TS	BHGA-SO	BHGA-ST	BHGA-SP
E22	best	384.67	384.67	384.67	384.67	384.67	384.67	384.67	384.67	384.67
	mean		384.67 ↓	384.67 ↓	384.67 ↓	384.67 ↓	384.67 ↓	384.67 ↓	384.67 ↓	384.67
	std.		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E23	best	571.94	571.94	571.94	571.94	571.94	571.94	571.94	571.94	571.94
	mean		571.94 ↓	571.94 ↓	571.94 ↓	571.94 ↓	571.94 ↓	571.94 ↓	571.94 ↓	571.94
	std.		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E30	best	509.47	509.47	509.47	509.47	509.47	509.47	509.47	509.47	509.47
	mean		509.47 ↓	509.47 ↓	509.47 ↓	509.47 ↓	509.47 ↓	509.47 ↓	509.47 ↓	509.47
	std.		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E33	best	840.14	844.25	840.14	840.57	840.57	840.14	840.14	840.14	840.14
	mean		845.62 ↑	840.43 ↑	842.30 ↑	843.59 ↑	840.70 ↑	840.14 ↓	840.14 ↓	840.14
	std.		0.92	1.18	1.42	1.29	1.40	0.00	0.00	0.00
E51	best	529.90	529.90	529.90	529.90	529.90	529.90	529.90	529.90	529.90
	mean		542.08 ↑	543.26 ↑	529.90 ↓	529.90 ↓	536.98 ↑	529.90 ↓	529.90 ↓	529.90
	std.		8.57	3.52	0.00	0.00	7.27	0.00	0.00	0.00
E76	best	692.64	697.27	692.64	692.64	692.64	692.74	692.64	692.64	692.64
	mean		717.30 ↑	697.89 ↑	692.85 ↑	694.58 ↑	694.96 ↑	692.64 ↓	692.64 ↓	692.64
	std.		9.58	3.09	0.81	3.33	1.63	0.00	0.00	0.00
E101	best	835.63	852.69	839.29	840.25	840.69	837.10	835.88	834.22	834.22
	mean		872.69 ↑	853.34 ↑	845.95 ↑	845.95 ↑	843.10 ↑	837.76 ↑	834.22 ↓	836.30
	std.		9.58	4.73	4.58	4.03	3.90	1.18	0.00	1.60
X143	best	15884.58	16488.60	16028.05	15901.23	15895.52	15910.86	15814.52	15813.71	15813.71
	mean		16911.50 ↑	16459.31 ↑	16031.46 ↑	15993.34 ↑	16214.37 ↑	15814.52 ↑	15814.41 ↓	15813.71
	std.		282.30	242.59	262.47	47.27	215.77	0.00	1.43	0.00
X214	best	11090.28	11762.07	11323.56	11133.14	11091.37	11090.28	10976.85	10976.92	10976.85
	mean		12007.06 ↑	11482.20 ↑	11219.70 ↑	11245.18 ↑	11206.60 ↑	10985.59 ↓	10988.73 ↓	10983.10
	std.		156.69	76.14	46.25	66.09	84.58	11.56	14.55	8.98
X351	best	26456.95	28008.09	27064.88	26478.34	26456.95	26622.42	26059.60	26067.16	26042.13
	mean		28336.07 ↑	27217.77 ↑	26593.18 ↑	26637.80 ↑	26750.60 ↑	26087.18 ↓	26134.58 ↑	26083.23
	std.		205.29	86.20	72.86	91.24	102.55	22.00	33.51	29.93
X459	best	24752.03	26048.21	25370.80	24763.93	24776.44	24794.35	24251.11	24248.04	24251.11
	mean		26345.12 ↑	25582.27 ↑	24916.60 ↑	24911.98 ↑	25041.10 ↑	24264.70 ↓	24279.36 ↑	24264.05
	std.		185.14	106.89	94.08	72.95	237.58	15.05	24.28	6.85
X573	best	51436.00	54189.62	52181.51	53822.87	53818.40	51436.90	50833.08	50917.26	50826.62
	mean		55327.62 ↑	52548.09 ↑	54567.15 ↑	54264.16 ↑	51776.70 ↑	50871.79 ↓	50975.03 ↑	50867.56
	std.		548.05	278.85	231.05	197.71	166.86	31.62	35.73	25.36
X685	best	69955.95	73925.56	71345.40	70834.88	70943.23	69955.95	68561.83	68730.36	68550.08
	mean		74508.03 ↑	71770.57 ↑	71440.57 ↑	71339.17 ↑	70401.25 ↑	68700.98 ↓	68855.52 ↑	68696.65
	std.		409.43	197.08	281.78	297.11	218.98	84.74	47.86	78.49
X749	best	79732.99	84034.73	81002.01	80299.76	80042.38	79779.87	77943.79	78179.01	78028.03
	mean		84759.79 ↑	81327.39 ↑	80694.54 ↑	80571.38 ↑	80135.67 ↑	78142.52 ↓	78403.74 ↑	78153.59
	std.		376.10	176.19	223.91	222.72	219.50	82.54	105.50	75.93
X819	best	161924.79	170965.68	164289.95	164720.80	163751.23	161924.79	158639.65	159031.50	158626.26
	mean		172410.12 ↑	164926.41 ↑	165565.79 ↑	164826.42 ↑	162530.67 ↑	158824.30 ↓	159455.62 ↑	158855.25
	std.		568.58	318.62	401.02	599.22	289.41	99.19	145.59	113.10
X916	best	336717.71	357391.57	341649.91	342993.01	341369.36	336717.71	330874.05	331618.09	330922.27
	mean		360269.94 ↑	342460.70 ↑	344999.95 ↑	343127.47 ↑	337641.92 ↑	331150.82 ↓	331931.77 ↑	331206.57
	std.		229.19	510.66	905.72	877.97	461.47	136.96	205.46	141.35
X1001	best	75469.29	78832.90	77476.36	76297.09	75666.26	75469.29	72885.35	73082.53	72954.16
	mean		79163.34 ↑	77920.52 ↑	77434.33 ↑	76405.53 ↑	75931.28 ↑	73064.77 ↓	73431.73 ↑	73154.48
	std.		229.19	234.73	719.8671	337.78	304.10	98.51	162.26	92.67
w/t/l			14/3/0	14/3/0	13/4/0	13/4/0	14/3/0	2/13/2	8/8/1	

B. Algorithm Settings

BHGA is implemented in C++ and runs on an AMD EPYC 7H12 2.6GHz processor. Each instance is tested 30 times with different random seed numbers to obtain reliable statistical results. The iterative stability coefficient $It_{\max}$ is set to 30,000. Other parameters are set following canonical HGS [18].

To better investigate the step parameter of AS, we designed three variants of BHGA: 1) BHGA-SO with step = 0; 2) BHGA-ST with step = 10, which totally uses AS; 3) BHGA-SP with step = 6, which partially uses AS. The reason for step = 6 is that the first six belong to basic moves, with

weaker selectivity. In general, BHGA refers to BHGA-SP.

In CEVRP benchmark instances, we compared these variants with GSGA, VNS, BACO, CBACO-I, and HHASA-TS. GSGA and VNS ranked as elite algorithms in the WCCI2020 competition of EVRP. BACO was prominent for its efficient bilevel optimization. CBACO and HHASA-RL updated multiple best known solutions, becoming state-of-the-art algorithms for solving CEVRP. For a fair comparison, we chose the best-performing variant of these two algorithms, namely CBACO-I and HHASA-TS.

Table II details the performance comparison of these al-

gorithms on benchmark instances, including instance name, minimum value, average value, and standard deviation. The results of these five algorithms are taken from [21], [16], [17]. BHGA-SO and BHGA-ST are compared with BHGA-SP using the Wilcoxon rank-sum test at a significance level of 0.05. Additionally, we performed a one-sample t-test comparing BHGA-SP to the remaining algorithms. The $\uparrow$, $\downarrow$, and $\downarrow$ means BHGA-SP is significantly better than, equal to, or significantly worse than the compared algorithm. w/t/l means on how many instances, BHGA-SP is better than, equal to, or worse than other algorithms.

VI. COMPARISONS AND ANALYSES

A. Investigation of Advanced Screening

The experimental results are shown in Table II. Through the comparison between BHGA-SO and BHGA-SP, we found AS to be extremely effective. In small-scale instances, this strategy helps population quickly converge to the best solution. Particularly in E101, both BHGA-SO and BHGA-SP successfully updated the best known solution. In large-scale instances, BHGA-SP becomes a more suitable choice. From X143 to X685, BHGA-SP performed better compared to BHGA-ST. In X749-X1001, the solution quality of BHGA also showed strong competitiveness, only slightly inferior to BHGA-SO.

It was observed that in X214-X1001, BHGA-ST did not surpass other variants. In practice, we found that the population does not need to evolve completely towards the direction of CEVRP. Overuse of AS might lead to early convergence to local optima, especially in large-scale instances. Consequently, maintaining the diversity of suboptimal solutions in the upper level CVRP is very crucial for exploring the best solution of CEVRP. Therefore, selecting the appropriate step of AS emerges as a vital consideration. BHGA-SP applies AS only to the first six weaker selective foundational moves, achieving a suitable balance between maintaining population diversity and finding the best solution.

B. Comparisons on Small-Scale Instances

According to the minimum object value in Table II, BHGA-SP and BHGA-ST found the best known solutions in all small-scale instances. From the perspective of the mean object value and standard deviation, BHGA shows reliable stability. To demonstrate, BHGA-ST achieved a zero standard deviation in all seven instances, and BHGA-SP also recorded a zero standard deviation in the first six instances.

Compared with other algorithms, BHGA exhibits more outstanding performance in small-scale instances. As an illustration, GSGA began to show a decrease in performance from E33 onwards, failing to identify the known best solutions in E33, E76, and E101, with significant fluctuations in objective values. Particularly in the instance E101 with the maximum number of customers, BHGA shows a higher solution quality compared to VNS, BACO, CBACO-I, and HHSA-TS.

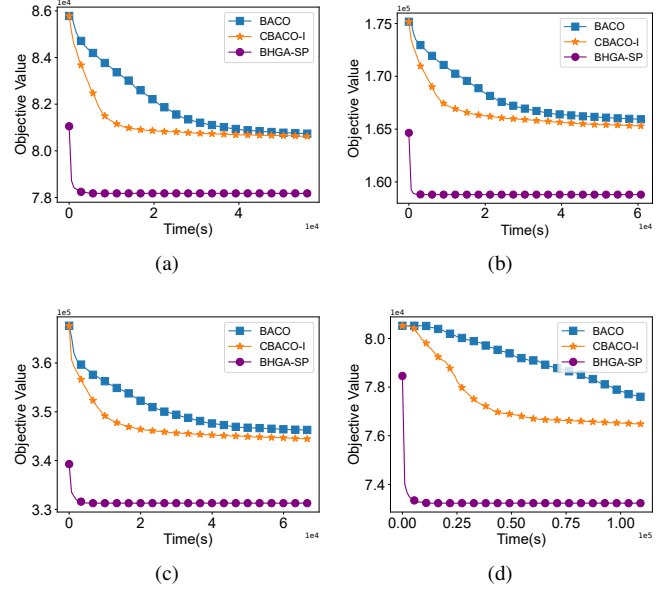


Fig. 2. Convergence curves of BACO, CBACO-I, and BHGA-SP. (a)-(d) represent X749-X1001. X-axis represents CPU time. Y-axis represents the objective value.

C. Comparisons on Large-Scale Instances

In terms of the minimum objective value in large-scale instances, BHGA successfully updated the best known solutions for all ten instances. In other statistical measures, the average solution quality of BHGA is the most outstanding. For example, in X1001, average solution quality of BHGA leads other algorithms, and it also achieves the minimum standard deviation.

Compared with other algorithms, BHGA has shown considerable superiority in handling large-scale instances. CBACO-I and HHSA-TS are state-of-the-art algorithms for CEVRP. Historically, in X143-X459, CBACO-I emerged as a superior solver, only slightly behind HHSA-TS in X214. In X573-X1001, HHSA-TS proved to be the appropriate choice, updating and setting new benchmarks for CEVRP. As of now, compared with these two algorithms, BHGA performed more effectively and stably, achieving further breakthroughs across the entire range of instances from X143 to X1001.

The efficiency of BHGA is due to its well-configured local search with AS and FE. The original local search move can efficiently explore the solution space, while AS further assists local search, guiding the population towards the evolution of CEVRP. FE can swiftly configure charging plans for electricity-infeasible routes, which is not only suitable for FRVCP but also beneficial to AS. In conclusion, through the close collaboration of the upper and lower subproblems, BHGA demonstrates strong competitiveness, becoming an advanced algorithm for CEVRP.

D. Convergence Analysis

In this section, we particularly focus on the large-scale instances ranging from X749 to X1001. The convergence behaviors of three algorithms BACO, CBACO-I, and BHGA

are illustrated in Fig. 2. To ensure effective convergence for BACO and CBACO-I, we set their execution time according to the parameters described in [16].

From these comparative analyses, it is evident that BHGA exhibited the most rapid convergence among all the evaluated algorithms in Fig. 2. By comparison, the two ACO-based bilevel algorithms BACO and CBACO-I, displayed stable performance in processing large-scale instances, yet their convergence rates were comparatively slower. In X749-X1001, BHGA demonstrated exceptional execution efficiency, achieving convergence to the best solution in an impressively brief duration. This efficiency is partly attributed to the application of local search with AS in the population initialization phase, which afforded BHGA a more advantageous starting position. The ability of FE to swiftly resolve FRVCP and generate high-quality solutions significantly advances the success of BHGA.

VII. CONCLUSION AND FUTURE WORKS

In this paper, we introduce a new algorithm named BHGA, specifically designed for efficiently solving CEVRP. BHGA exhibits performance surpassing state-of-the-art algorithms, updating eleven out of seventeen best known solutions in the CEVRP benchmark instances. Specifically, BHGA achieves more effective solutions by dividing CEVRP into two levels of subproblem: 1) CVRP and 2) FRVCP. In handling CVRP, we applied AS to the local search part of HGS to obtain better CVRP routes. In terms of FRVCP, The FE heuristic can efficiently construct charging plans for CVRP routes. FE not only improves the quality of charging plans but also aids in the evaluation of AS in the upper level, realizing an effective coupling between CVRP and FRVCP. In terms of solution quality, BHGA completely surpasses the CBACO and HHSA-RL algorithms in large-scale benchmark instances. Concerning time efficiency, BHGA shows a faster convergence speed compared to CBACO and BACO, demonstrating a clear advantage. In future research, we will explore applying BHGA to EVRPs like EVRPTW and EVRP with pickup and delivery.

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